

Solution Requirements

Date	11 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users can securely access the HDI Prediction System using a username and password (optional for project demo).</i>	High
2	Authorization levels	<i>Admin can manage datasets and models. Users can enter indicators and view HDI predictions.</i>	High

3	External interfaces	<i>The system uses HDI datasets from UNDP, World Bank, or Kaggle and interacts with the Flask web interface.</i>	High
4	Transactions processing	<i>The system accepts user inputs (Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, GNI), processes them through the ML model, and returns the predicted HDI category.</i>	High
5	Reporting	<i>Displays the predicted HDI category and allows viewing model accuracy, confusion matrix, and classification report.</i>	Medium
6	Business rules, etc.	<i>The prediction is based only on valid numeric input values. The trained Machine Learning model classifies countries into Very High, High, Medium, or Low HDI.</i>	High
7	Compliance to laws or regulations	<i>Uses publicly available datasets from trusted sources (UNDP/World Bank/Kaggle) and does not store personal user information.</i>	Medium

8	Other	<i>Provides a simple web interface for entering country indicators and viewing prediction results.</i>	Medium
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Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	<i>The prediction result should be generated quickly after user input.</i>	<i>Response time less than 2 seconds)</i>
2	Scalability	<i>The system should support multiple users and larger datasets in the future.</i>	<i>Supports 100+ prediction requests without performance issues</i>
3	Security & Data Privacy	<i>Input data should be validated, and no sensitive user information should be stored.</i>	<i>Secure input validation and protection against invalid data)</i>
4	Reliability & Availability	<i>The system should be available whenever users access it and produce consistent predictions.</i>	<i>99% system availability</i>
5	Usability & Accessibility	<i>The web interface should be simple, responsive, and easy for students, researchers, and policymakers to use.</i>	<i>Easy navigation with a user-friendly interface</i>

6	<i>Other</i>	<i>The system should be easy to update with new datasets or improved ML models and should run on Windows, Linux, and cloud platforms.</i>	Modular code with easy deployment and maintenance
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